University of Zurich[UZH]

Department of Informatics

Graph Neural Networks for Epidemic Source Detection on Temporal Contact Networks

Master Thesis Proposal

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Introduction

Identifying the origin of an epidemic, often termed "patient zero," is a fundamental challenge in network science. The increasing availability of high-resolution contact data ranging from Bluetooth proximity sensors to industrial animal movements enabled the application of computational methods to this problem [15]. Beyond biological pathogens, these methods are also applicable to other domains, such as identifying the origin of information spreading on social media [4].

Early research focused on static networks, introducing rumor centrality as a maximum likelihood estimator that identifies the source by analyzing the combinatorial number of permitted infection sequences [14]. However, temporal networks provide a much more realistic substrate for modeling epidemics by accounting for the precise timing and order of contacts. This realization has sparked significant research interest in temporal source detection, leading to methods such as individual-based approximations [9], dynamic message-passing [5, 6], soft margin estimator [1], loopy belief propagation [3], and Steiner-tree approximations [10].

More recently, Graph Neural Networks (GNNs) have emerged as a powerful alternative by treating source detection as a supervised learning task. GNNs are particularly well-suited for this domain because their message-passing mechanism aggregates neighborhood information to determine whether a node is a likely source based on local infection densities [4]. By training on large-scale simulations of epidemic outcomes, GNNs can learn complex structural patterns that conventional heuristics fail to capture [13].

Despite their potential, most GNN-based architectures are restricted to the static projection of networks, thereby ignoring the critical temporal dimension of real-world interactions. While initial efforts have been made to adapt GNNs to temporal networks [7, 11], there is currently no systematic evaluation of their performance or their ability to outperform static baselines. This thesis seeks to bridge this gap by developing and evaluating GNN architectures specifically designed for temporal epidemic source detection.

Formalized Problem

The source detection problem is to identify the origin of an epidemic spreading among humans, animals, or other actors. Formally, we are given a temporal contact network $G(V, E^t)$, where each edge $(u, v, t) \in E^t$ represents a contact between nodes u and v at time t, and a snapshot of the epidemic: the states of some or all nodes at a fixed time t_{end} after the epidemic started at time t_0. The goal is to infer a probability distribution (at a certain time) over all possible source nodes $v \in V$.

Training data is generated by simulating epidemic outbreaks using the Susceptible-Infectious (SI) or Susceptible-Infectious-Recovered (SIR) models. This generation relies on a Monte Carlo process, a stochastic simulation technique that uses repeated random sampling to model the probabilistic nature of disease transmission. Specifically, at each time step, an infected node transmits the pathogen to a susceptible neighbor with a transmission probability β, while in the SIR model, infected nodes additionally transition to a recovered state with a recovery probability μ. Ground-truth snapshots used for model evaluation are generated using the same underlying SI/SIR models and Monte Carlo processes.

Task Description

This master's thesis consists of the following three tasks:

Sub-task 1. Literature Review and Conceptualization

Establish a theoretical foundation for epidemic source detection and GNNs.
- Familiarize yourself with common models of epidemic spreading, such as the SI and SIS models.
- Conduct a review of classical source detection algorithms on temporal networks.
- Learn the foundations of GNNs.
- Review current state-of-the-art GNN approaches for static and temporal networks.

Sub-task 2. Implementation of Temporal GNN Architectures

Implement and adapt various GNN frameworks for the temporal case using PyTorch and PyTorch Geometric framework.
- Implement a baseline GNN using a static projection of the temporal network.
- Implement a primitive temporal GNN where each layer corresponds to a timeslice of the network.
- Implement the de Bruijn GNN architecture based on recent work by Qarkaxhija et al. [7]
- Implement a Kernel-based GNN approach as proposed by Ru et al. [11]
- Implement a DAG-GNN by Rey et al. [8] on the temporal event graph [12]
- Others as appropriate.

Sub-task 3. Evaluation and Comparative Analysis

Systematically evaluate the performance of the implemented models.
- Construct temporal contact networks from synthetic graphs or real-world data.
- Apply the implemented GNN models for source detection on these networks.
- Systematically evaluate and quantify the performance gains of temporal-aware models compared to static baselines.
- Perform experiments to identify scenarios where specific models outperform or underperform, e.g., depending on network density.

General Thesis Guidelines

All source code written as a part of this thesis should be released under suitable open-source licenses. The typical rules of academic work must be followed. Consider the guidelines described in [2]. At the end of the thesis, a final report (thesis) has to be written. The report should be clearly organized and follows the usual academic report structure. A LaTeX template will be provided upon request.

Tutor

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Signatures

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References

[1] N. Antulov-Fantulin, A. Lancic, T. Smuc, H. Stefancic, and M. Sikic. Identification of Patient Zero in Static and Temporal Networks - Robustness and Limitations. *Physical Review Letters*, 114(24):248701, June 2015. arXiv:1406.2909 [cs].

[2] A. Bernstein. So what is a (Diploma) Thesis? A few thoughts for first-timers., 2005.

[3] A. Braunstein and A. Ingrosso. Inference of causality in epidemics on temporal contact networks. *Scientific Reports*, 6(1), June 2016. Publisher: Springer Science and Business Media LLC.

[4] M. Dong, B. Zheng, N. Quoc Viet Hung, H. Su, and G. Li. Multiple Rumor Source Detection with Graph Convolutional Networks. In *Proceedings of the 28th ACM International Conference on Information and Knowledge Management*, pages 569–578, Beijing China, Nov. 2019. ACM.

[5] A. Koher, H. H. Lentz, J. P. Gleeson, and P. Hövel. Contact-Based Model for Epidemic Spreading on Temporal Networks. *Physical Review X*, 9(3):031017, Aug. 2019.

[6] A. Y. Lokhov, M. Mézard, H. Ohta, and L. Zdeborová. Inferring the origin of an epidemic with a dynamic message-passing algorithm. *Physical Review E*, 90(1), July 2014. arXiv:1303.5315 [physics].

[7] L. Qarkaxhija, V. Perri, and I. Scholtes. De Bruijn goes Neural: Causality-Aware Graph Neural Networks for Time Series Data on Dynamic Graphs, Sept. 2022. arXiv:2209.08311 [cs].

[8] S. Rey, H. Ajorlou, and G. Mateos. Directed Acyclic Graph Convolutional Networks, June 2025. arXiv:2506.12218 [eess].

[9] L. E. C. Rocha and N. Masuda. Individual-based approach to epidemic processes on arbitrary dynamic contact networks. *Scientific Reports*, 6(1), Aug. 2016. Publisher: Springer Science and Business Media LLC.

[10] P. Rozenshtein, A. Gionis, B. A. Prakash, and J. Vreeken. Reconstructing an Epidemic Over Time. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 1835–1844, San Francisco California USA, Aug. 2016. ACM.

[11] X. Ru, J. Murdoch Moore, X.-Y. Zhang, Y. Zeng, and G. Yan. Inferring Patient Zero on Temporal Networks via Graph Neural Networks. *Proceedings of the AAAI Conference on Artificial Intelligence*, 37(8):9632–9640, June 2023. Publisher: Association for the Advancement of Artificial Intelligence (AAAI).

[12] J. Saramäki, M. Kivelä, and M. Karsai. Weighted temporal event graphs, Dec. 2019. arXiv:1912.03904 [physics].

[13] C. Shah, N. Dehmamy, N. Perra, M. Chinazzi, A.-L. Barabási, A. Vespignani, and R. Yu. Finding Patient Zero: Learning Contagion Source with Graph Neural Networks, June 2020. arXiv:2006.11913 [cs].

[14] D. Shah and T. Zaman. Detecting sources of computer viruses in networks: theory and experiment.

[15] M. Sterchi, L. Hilfiker, R. Grütter, and A. Bernstein. Active querying approach to epidemic source detection on contact networks. *Scientific Reports*, 13(1), July 2023. Publisher: Springer Science and Business Media LLC.